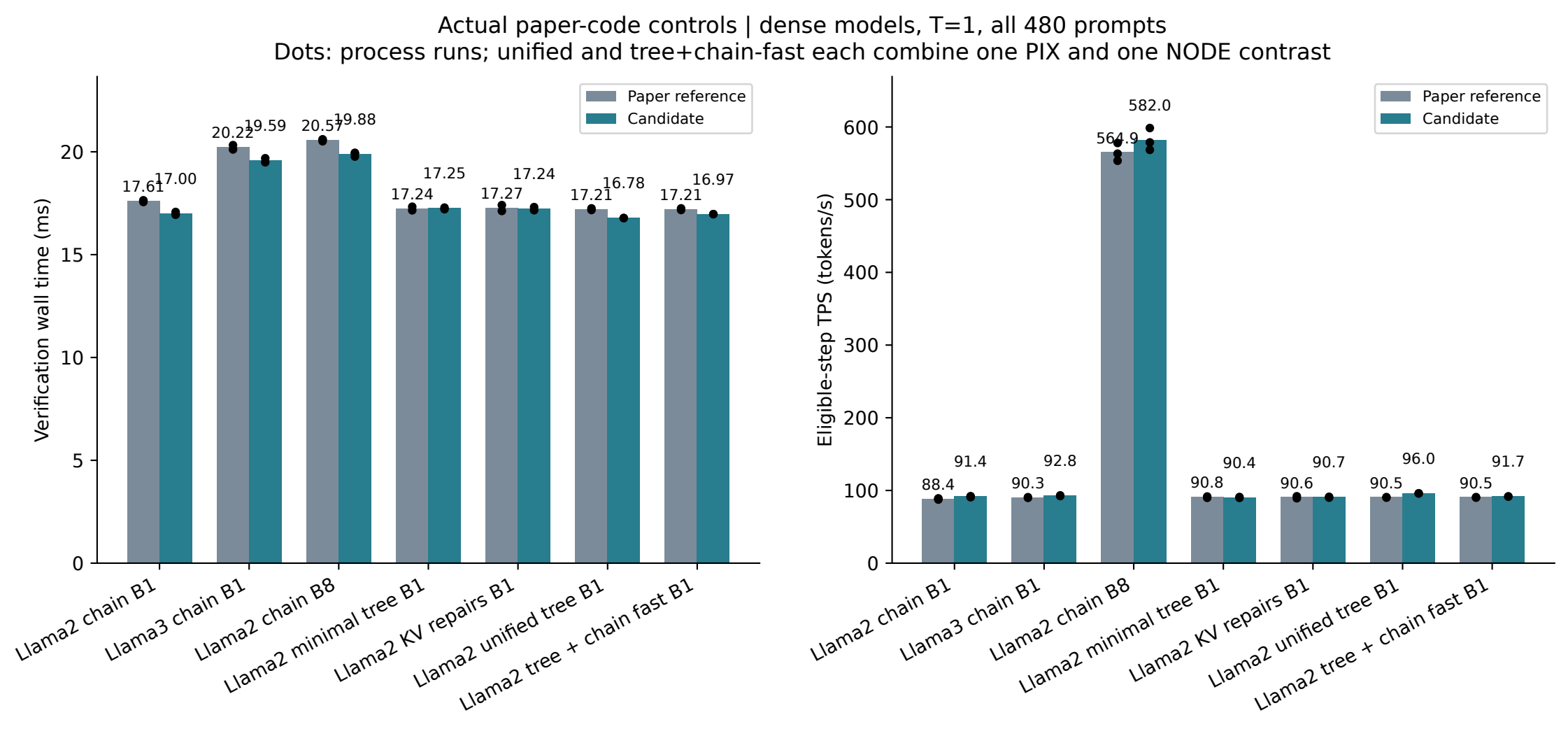
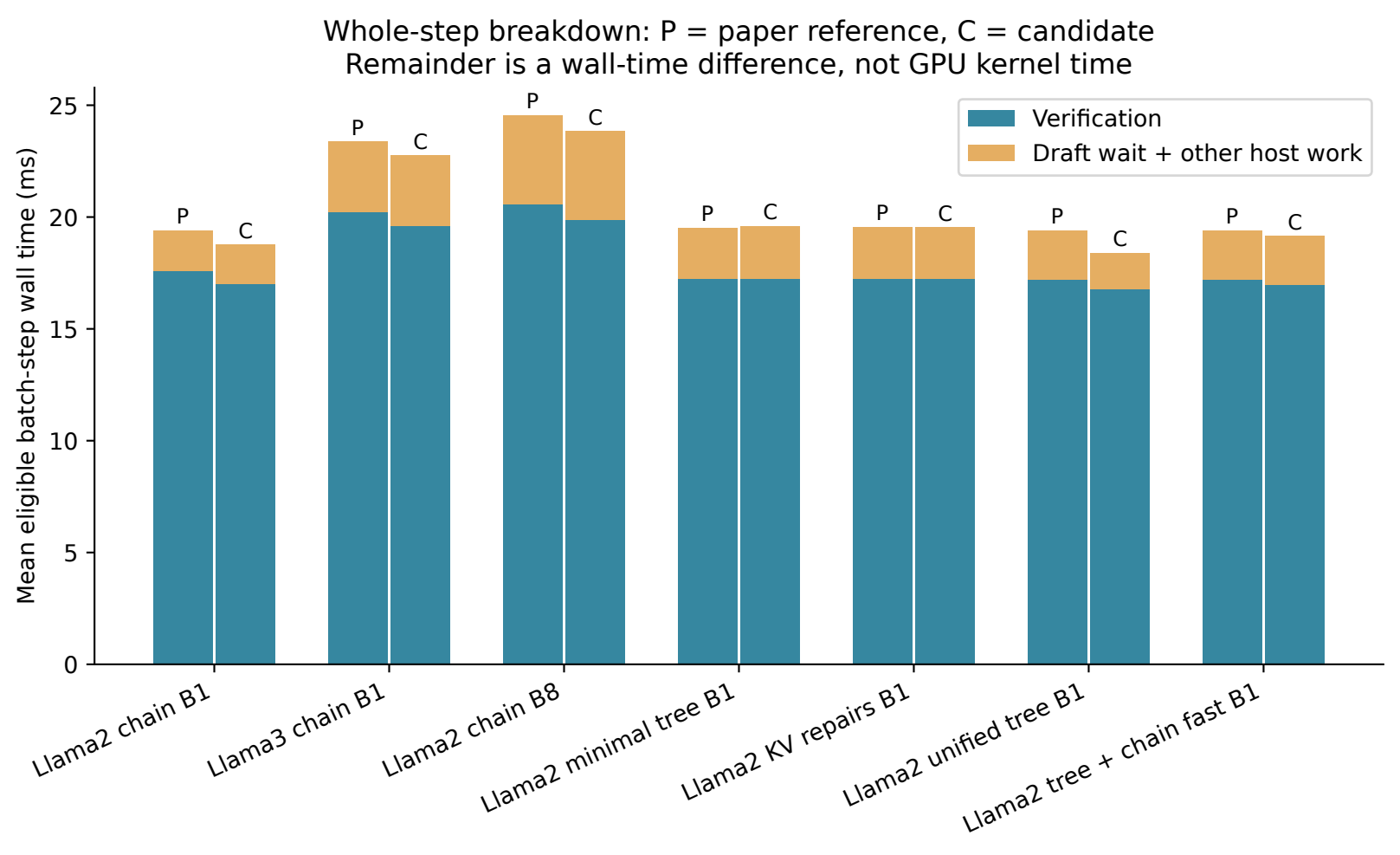
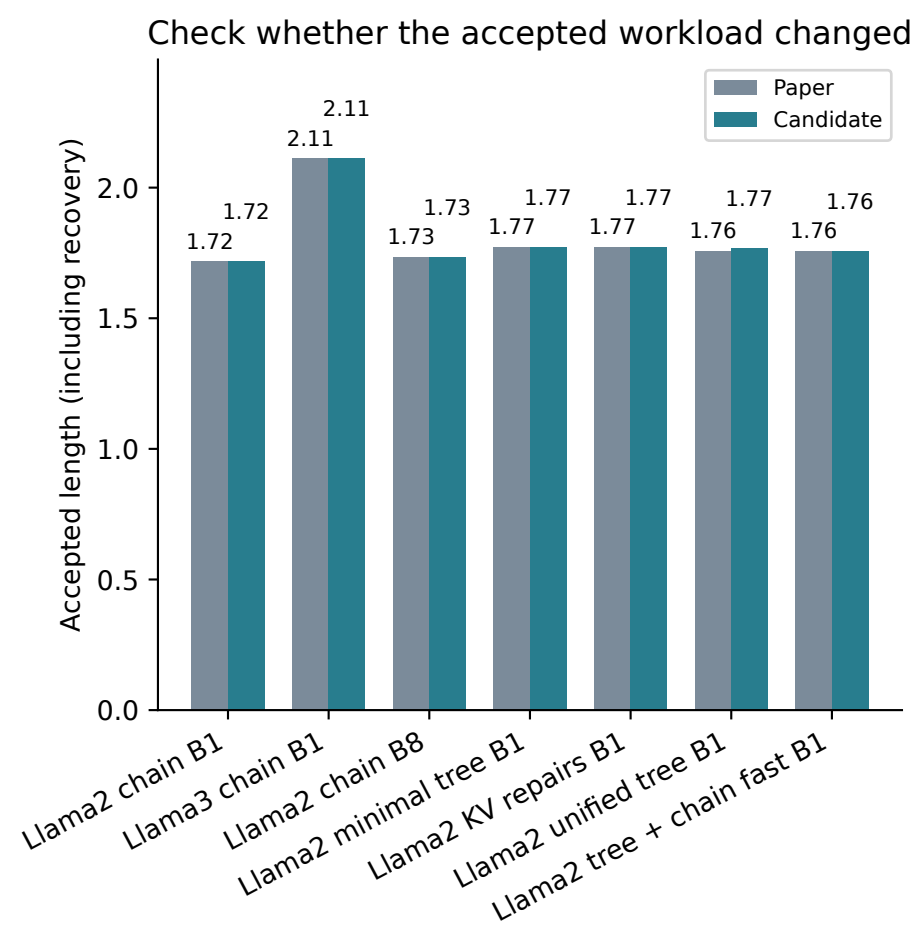
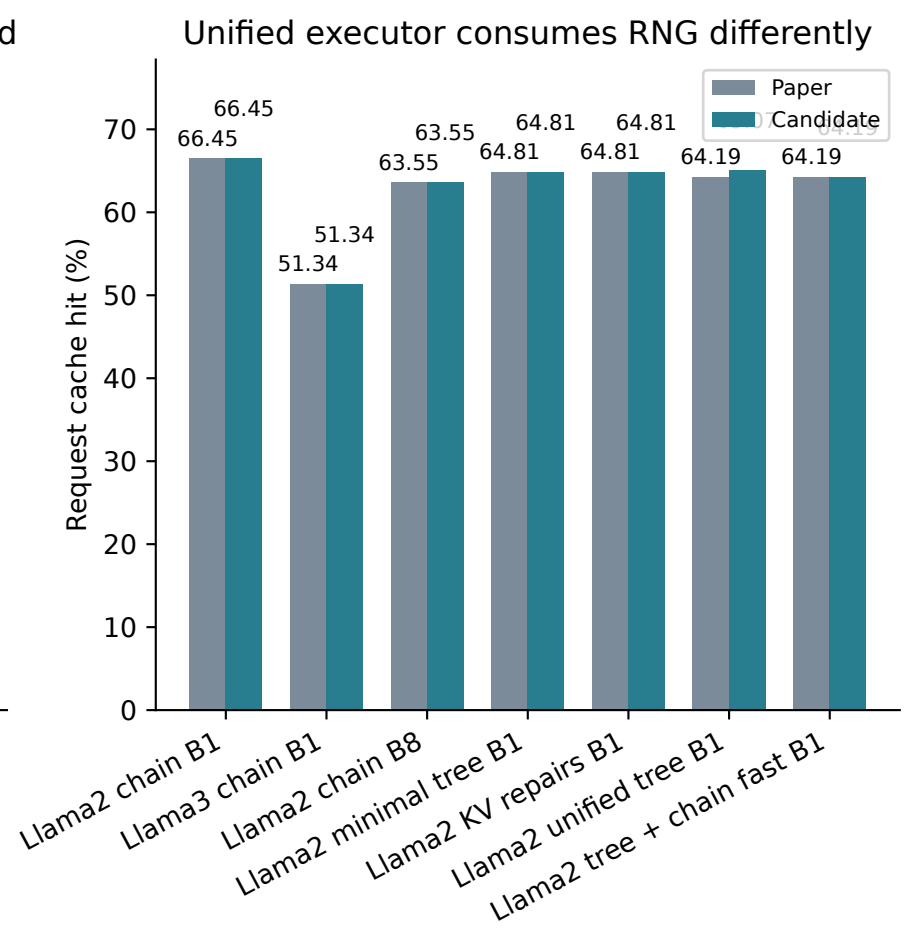
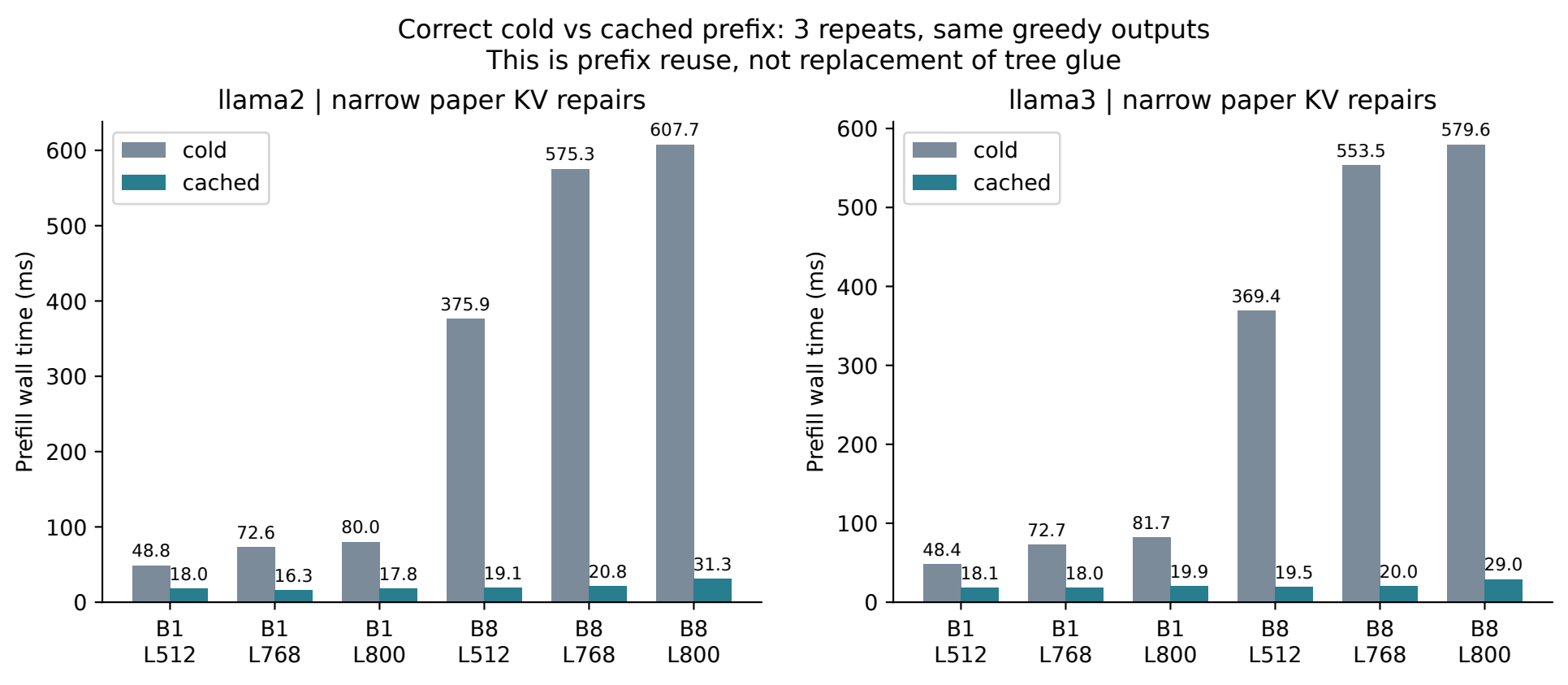
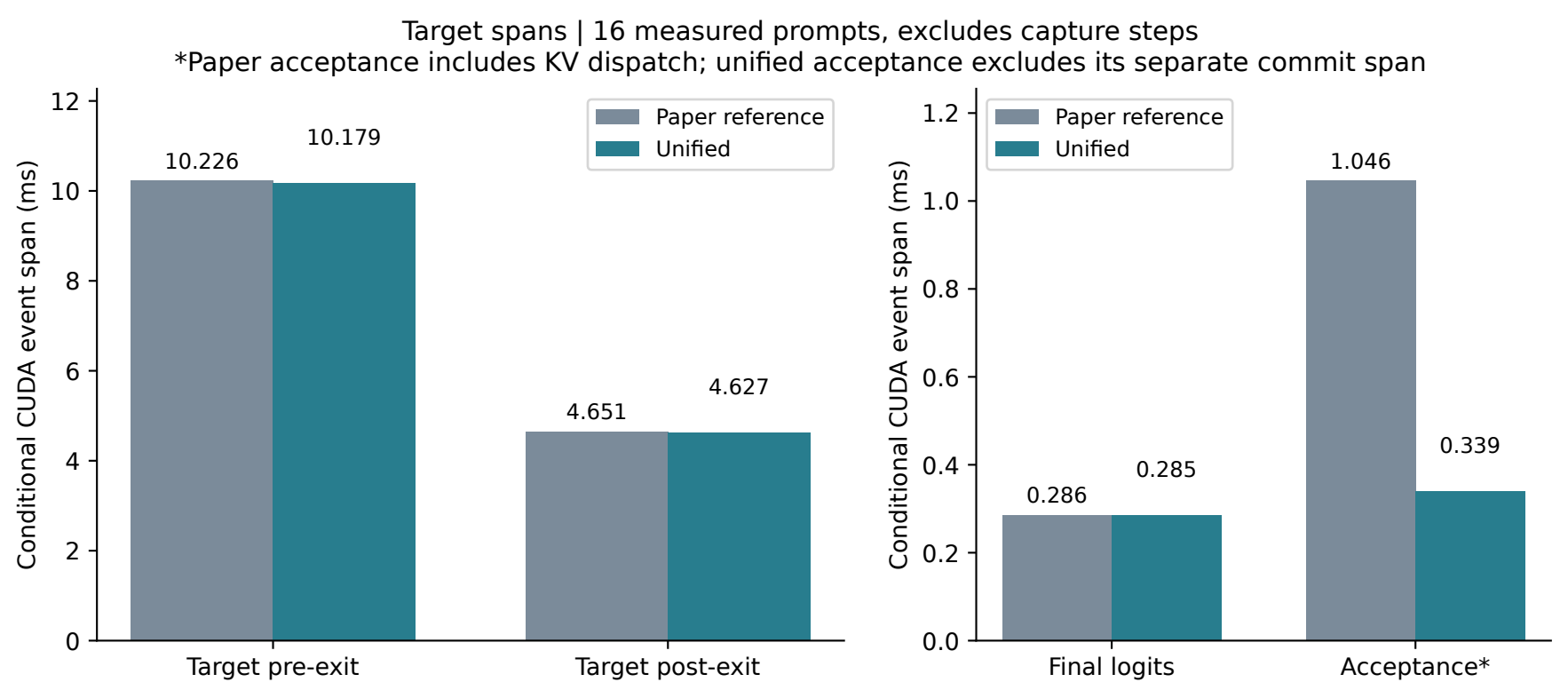

Actual paper-code controls | dense models, T=1, all 480 prompts
Dots: process runs; unified and tree+chain-fast each combine one PIX and one NODE contrast
Paper reference
Candidate
Verification wall time (ms)
0
5
10
15
20
17.61
17.00
20.22
19.59
20.57
19.88
17.24
17.25
17.27
17.24
17.21
16.78
17.21
16.97
Eligible-step TPS (tokens/s)
0
100
200
300
400
500
600
88.4
91.4
90.3
92.8
564.9
582.0
90.8
90.4
90.6
90.7
90.5
96.0
90.5
91.7
Llama2 chain B1
Llama3 chain B1
Llama2 chain B8
Llama2 minimal tree B1
Llama2 KV repairs B1
Llama2 unified tree B1
Llama2 tree + chain fast B1


Whole-step breakdown: P = paper reference, C = candidate
Remainder is a wall-time difference, not GPU kernel time
Mean eligible batch-step wall time (ms)
0
5
10
15
20
25
Verification
Draft wait + other host work
P
C
Llama2 chain B1
Llama3 chain B1
Llama2 chain B8
Llama2 minimal tree B1
Llama2 KV repairs B1
Llama2 unified tree B1
Llama2 tree + chain fast B1


Check whether the accepted workload changed
Paper
Candidate
Accepted length (including recovery)
0.0
0.5
1.0
1.5
2.0
1.72
1.72
2.11
2.11
1.73
1.73
1.77
1.77
1.77
1.77
1.76
1.77
1.76
1.76
Llama2 chain B1
Llama3 chain B1
Llama2 chain B8
Llama2 minimal tree B1
Llama2 KV repairs B1
Llama2 unified tree B1
Llama2 tree + chain fast B1


Unified executor consumes RNG differently
Paper
Candidate
Request cache hit (%)
0
10
20
30
40
50
60
70
66.45
66.45
51.34
51.34
63.55
63.55
64.81
64.81
64.81
64.81
64.19
64.19
Llama2 chain B1
Llama3 chain B1
Llama2 chain B8
Llama2 minimal tree B1
Llama2 KV repairs B1
Llama2 unified tree B1
Llama2 tree + chain fast B1


Correct cold vs cached prefix: 3 repeats, same greedy outputs
This is prefix reuse, not replacement of tree glue
llama2 | narrow paper KV repairs
llama3 | narrow paper KV repairs
cold
cached
Prefill wall time (ms)
0
100
200
300
400
500
600
48.8
18.0
72.6
16.3
80.0
17.8
375.9
19.1
575.3
20.8
607.7
31.3
48.4
18.1
72.7
18.0
81.7
19.9
369.4
19.5
553.5
20.0
579.6
29.0
B1 L512
B1 L768
B1 L800
B8 L512
B8 L768
B8 L800


Target spans | 16 measured prompts, excludes capture steps
*Paper acceptance includes KV dispatch; unified acceptance excludes its separate commit span
Conditional CUDA event span (ms)
Paper reference
Unified
10.226
10.179
4.651
4.627
Target pre-exit
Target post-exit
0.286
0.285
1.046
0.339
Final logits
Acceptance*